Econometrics II – Assignment 2: Geopolitical Risk and Economic Activity

Spring 2026

1 Introduction

This paper analyzes the dynamic relationship between geopolitical risk, stock returns, and U.S. economic activity using a monthly VAR model. The central question is whether geopolitical risk drives financial and real activity dynamics, whether reverse predictive channels are present, and whether there is evidence of contemporaneous comovement.

The preferred model is a reduced-form VAR selected by information criteria and validated through misspecification diagnostics. Baseline Gaussian Wald tests show no Granger causality from geopolitical risk to either stock returns or production growth, and baseline evidence for $s \to y$. Because residual diagnostics indicate heteroskedasticity and non-normality, we report HAC-robust Wald tests as the main inferential benchmark. Under robust inference, the conclusion that geopolitical risk has weak predictive power is unchanged, while the $s \to y$ channel is no longer significant at 5%.

2 Description of the Data

The sample contains monthly observations from January 1985 to December 2025 (492 usable observations). The variables and transformations are:

$$\begin{aligned} g_t &= \log(GPR_t), \\ y_t &= \log(INDPRO_t) - \log(INDPRO_{t-1}), \\ s_t &= \log\left(1 + \frac{Mkt_t}{100}\right). \end{aligned}$$

Thus, y_t is monthly industrial production growth and s_t is continuously compounded monthly stock return.

Data sources are the Caldara-Iacoviello geopolitical risk database (GPR), FRED (INDPRO), and the Fama-French data library (Mkt).

Figure 1 displays the three series with major event windows (9/11, COVID-19, 2022 war shock, and 2023–2024 Middle East escalation). The data show large crisis outliers, especially in y_t during early 2020, and visibly non-Gaussian tail behavior.

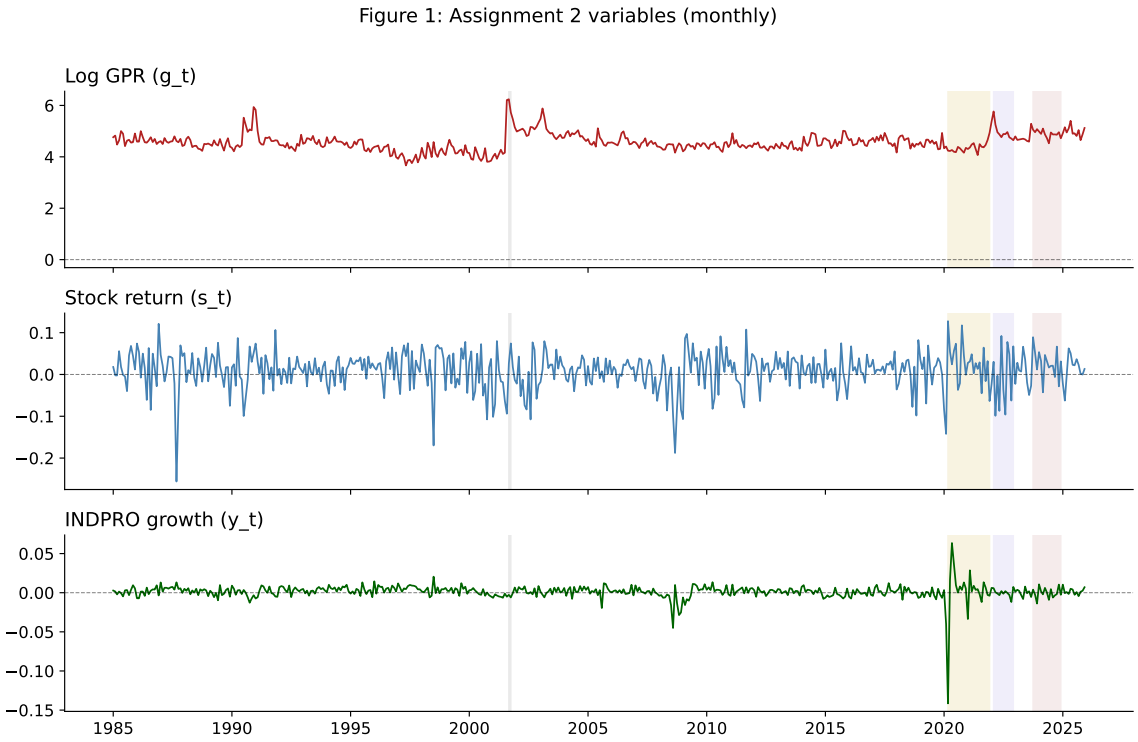


Figure 1: Monthly series for g_t, s_t, and y_t with highlighted event windows.

3 Econometric Theory

The assignment studies whether geopolitical risk helps predict stock returns and real activity, and whether the relationship is mainly lagged (predictive) or contemporaneous. To analyze this, we model the three monthly variables jointly as

$$Z_t = (g_t, s_t, y_t)',$$

where g_t is log geopolitical risk, s_t is log stock return, and y_t is industrial production growth.

The baseline model is a reduced-form VAR(k):

$$Z_t = \mu + \Pi_1 Z_{t-1} + \cdots + \Pi_k Z_{t-k} + e_t, \tag{1}$$

with innovations satisfying

$$E(e_t \mid \mathcal{I}_{t-1}) = 0, \qquad E(e_t e_t') = \Omega, \qquad E(e_t e_{t-j}') = 0 \text{ for } j \geq 1,$$

and Ω symmetric positive definite. Thus, each variable is explained by its own lags and lags of the other variables, while contemporaneous comovement is summarized by Ω.

A key requirement for standard VAR inference is stability (covariance stationarity). For a VAR(k), this is checked in companion form: all eigenvalues must lie strictly inside the unit circle (equivalently, all roots of the characteristic polynomial lie outside the unit circle). Under stability and standard regularity conditions, equation-by-equation OLS is numerically identical to Gaussian MLE in the VAR, and

$$\sqrt{T}\,\mathrm{vec}\Big(\hat{\Phi} - \Phi\Big) \xrightarrow{d} N(0, V),$$

so Wald and LR tests are asymptotically valid for linear restrictions.

To test predictive (Granger) causality, we use block exclusion restrictions. For example, "g does not Granger-cause s" in VAR(k) is

$$H_0 : \ \Pi_{s,g,1} = \Pi_{s,g,2} = \cdots = \Pi_{s,g,k} = 0, \tag{2}$$

i.e., all coefficients on g_{t-j} in the s_t-equation are jointly zero. Under correct specification, the Wald statistic is asymptotically $\chi^2(m)$, where m is the number of restrictions (here $m = k$). Granger causality is about incremental forecasting power and does *not* identify contemporaneous causality.

Misspecification affects inference: if residuals are heteroskedastic or non-Gaussian, conventional (model-based) covariance estimates can understate uncertainty. Following the course-note guidance, we therefore complement baseline Gaussian Wald tests with HAC-robust Wald tests for key restrictions and treat robust inference as primary whenever conclusions differ.

For dynamic interpretation of shocks, we report impulse-response functions (IRFs). Reduced-form shocks are generally correlated, so economically interpretable IRFs require orthogonalization. We use a recursive (Cholesky) identification with ordering (g, s, y), which imposes that geopolitical risk is predetermined within the month relative to stock returns and production growth. Hence, IRFs are interpreted conditionally on this maintained ordering assumption; in particular, contemporaneous feedback to g_t within the month is ruled out by construction.

4 Empirical Analysis

4.1 Modeling Progress and Model Selection

We estimate VAR(k) for $k = 1, 2, 3, 4$ and compare AIC, BIC, and HQIC. BIC selects VAR(1), while AIC is slightly lower at VAR(3). Because each extra lag adds nine slope coefficients in this three-variable system, we follow BIC and retain VAR(1) as preferred, then evaluate misspecification before inference. A short VAR(2) sensitivity check is discussed in the robustness paragraph. Table 1 summarizes lag-selection results.

Table 1: VAR lag-order selection.

	AIC	BIC	HQIC	FPE
1	-18.5525	-18.4499	-18.5122	0.0000
2	-18.5615	-18.3818	-18.4910	0.0000
3	-18.5772	-18.3200	-18.4762	0.0000
4	-18.5483	-18.2134	-18.4167	0.0000

4.2 Misspecification and Consequences for Inference

Equation-level diagnostics indicate residual autocorrelation in parts of the system and strong non-normality. Residual autocorrelation is borderline in the g_t equation and stronger in the y_t equation at lag 12. ARCH-LM rejects homoskedasticity for s_t and y_t, and Jarque-Bera rejects normality in all equations. Compact p-value diagnostics are shown in Table 2.

Table 2: Compact misspecification diagnostics (p-values) for preferred VAR(1).

	LB p	ARCH p	JB p	Any reject (5%)
g	0.0480	0.9636	0.0000	Yes
s	0.5401	0.0050	0.0000	Yes
y	0.0013	0.0000	0.0000	Yes

These outcomes imply that standard Gaussian Wald significance can be misleading in finite samples. Estimation of dynamic coefficients remains informative, but formal hypothesis testing should rely on robust covariance methods. We therefore treat HAC-based Wald tests as the main inferential results.

Contemporaneous dependence is assessed by residual correlations. The estimated values are moderate: $\mathrm{corr}(e_g, e_s) \approx -0.106$, $\mathrm{corr}(e_g, e_y) \approx 0.004$, and $\mathrm{corr}(e_s, e_y) \approx -0.058$.

These correlations motivate caution in structural interpretation, since recursive contemporaneous restrictions are not directly testable.

4.3 Predictive Channels: Baseline and Robust Tests

Baseline VAR Wald tests show no Granger-causality from g to s or y, and baseline evidence for $s \to y$.

Given misspecification evidence, we also test key restrictions using HAC covariance in equation-by-equation regressions. The robust tests confirm weak predictive content from g to both s and y. The $s \to y$ channel is no longer significant at 5% under HAC, which indicates that the baseline significance partly reflects non-constant volatility. Baseline and HAC p-values are summarized in Table 3.

Table 3: Baseline versus HAC p-values for key predictive restrictions.

	Baseline p	HAC p	Baseline reject 5%	HAC reject 5%
g -> s	0.4779	0.5601	No	No
g -> y	0.2484	0.1716	No	No
s -> y	0.0000	0.1613	Yes	No

4.4 Impulse Responses and Sample Robustness

Figure 2 shows orthogonalized IRFs from the preferred VAR(1). Under the maintained Cholesky ordering, a one-standard-deviation GPR shock reduces stock returns on impact (around -0.0048 in log-return units), then decays quickly. The response of production growth is small and close to zero.

Figure 3: Orthogonalized IRFs to a GPR shock (VAR preferred)

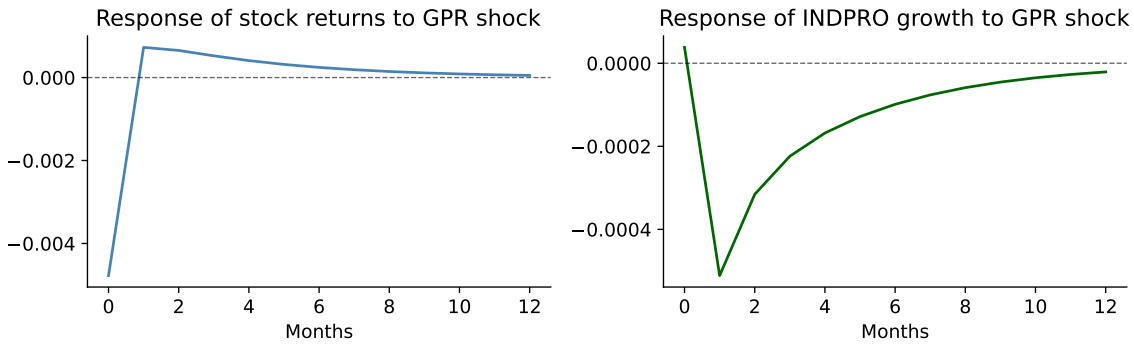


Figure 2: Orthogonalized impulse responses to a GPR shock from VAR(1) with Cholesky ordering (g, s, y).

We re-estimate the model on two alternative samples: post-9/11 only, and a sample

excluding 2020–2024; we also check VAR(2) briefly as a sensitivity analysis. The qualitative message is unchanged: $g \to s$ and $g \to y$ predictive channels remain weak (with only borderline evidence in one restricted sample), while financial dynamics continue to be the dominant short-run channel.

5 Discussion

The empirical strategy is transparent and closely aligned with the assignment objective of separating predictive dynamics from contemporaneous comovement in a multivariate setting. The preferred VAR(1) provides a clear and interpretable benchmark.
A key strength is consistency across methods: model selection, diagnostics, Granger-causality tests, and IRFs all point to the same qualitative conclusion. Lagged geopolitical risk has limited predictive power for stock returns and industrial production growth in this specification.

At the same time, three limitations are important. First, the three-variable VAR is intentionally parsimonious and cannot capture all macro-financial transmission channels emphasized in larger GPR systems; omitted variables may attenuate direct GPR effects. Second, misspecification tests indicate heteroskedasticity and non-normal residuals, especially during crisis periods. This weakens reliance on conventional Gaussian finite-sample p-values and motivates robust inference as the primary benchmark.
Third, orthogonalized IRFs rely on the maintained Cholesky ordering (g, s, y). Ordering GPR first is economically plausible at monthly frequency, but contemporaneous restrictions are not directly testable, so structural interpretation should remain cautious.

Overall, the evidence suggests that in this compact monthly system, financial variables are the main short-run transmission channel, while direct predictive effects of aggregate geopolitical risk on real activity are modest.

6 Conclusion

Using monthly U.S. data from 1985–2025, we estimate a parsimonious VAR for geopolitical risk, stock returns, and industrial production growth. Lag-order criteria point to a short dynamic structure, and we therefore use VAR(1) as the preferred specification.
Our main finding is that lagged geopolitical risk does not provide robust predictive content for either stock returns or production growth in this setup. This conclusion is stable across baseline Granger-type tests and HAC-robust inference, despite evidence of heteroskedasticity and non-normal residuals.
Baseline results indicate a predictive channel from stock returns to production growth,

but this channel is not significant at the 5% level once robust covariance estimation is applied. This difference between baseline and robust inference suggests that finite-sample significance can be sensitive to residual volatility patterns.

Orthogonalized impulse responses, interpreted conditional on the Cholesky ordering (g, s, y), show an immediate negative response of stock returns to a GPR shock that decays relatively quickly. The direct response of production growth is small and close to zero over the horizon considered, consistent with limited short-run real-activity transmission in this compact system.

Robustness checks based on alternative samples (post-9/11 and exclusion of 2020–2024) do not alter the qualitative conclusions. The overall picture is therefore that financial variables remain the dominant short-run adjustment margin, while direct predictive effects of aggregate geopolitical risk on real activity are modest.

Taken together, the evidence supports a cautious interpretation: geopolitical risk is economically relevant for short-run financial conditions, but its direct predictive role for monthly real activity is weak in this three-variable VAR framework.